

NC STATE UNIVERSITY

ECE 533 Final Project

**Development and Electrical Characterization of a
Shielded ½ Bridge SiC Power Module with Physical
Design**

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Problem 1: Electrical Analysis of a ½ Bridge IPM and DPT

Overview

Interstitial shield layer (or interstitial layer) reduces the offending current flowing to the ground. In the following LTSPICE simulations, we have compared the ground current (offending current) in case of circuit **without having interstitial layer (case A)** and **with interstitial layer (case B)**. In the equivalent electrical circuit, interstitial layer can be represented as a branch of impedance as series of parasitic resistance and inductance of the interstitial layer in the HBPM.

Double Pulse Test

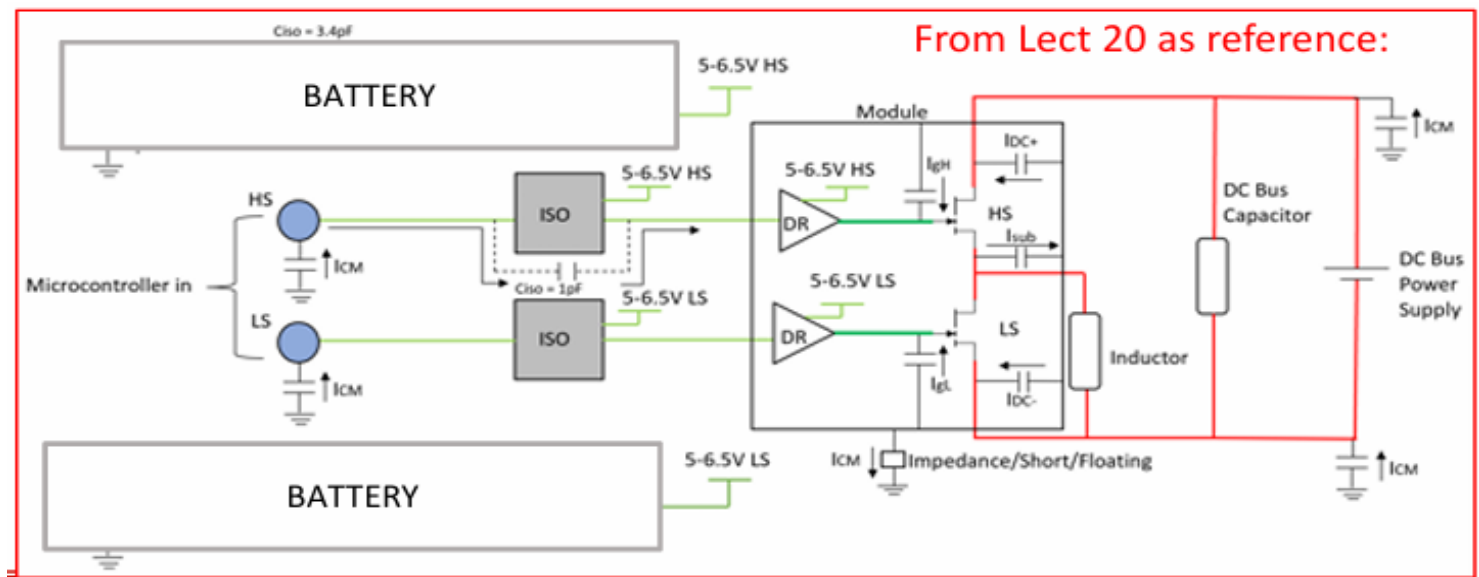


Fig1: DPT

Ground current (offending current) I_{gnd}

Current flowing through the ground branch ($L23+R9+gnd$ in Fig1) and ($C9+L11+R4+gnd$ in Fig2)

The ground current waveforms in the two cases A and B cannot be presented directly as it having only spikes. Instead, those currents are presented as Voltages(noise) in bode-plot for better visible analysis.

Bode-plots presented consists of decibel(dB) on Y-axis and Frequency (Hz) on X-axis: Fig3, Fig8 and Fig12

PROBLEM 1A

Case A) Without interstitial layer

Gate pulse: PWL (0 0 4.99u 0 5u 20 19.99u 20 20u 0 24.99u 0 25u 20 39.99u 20 40u 0)

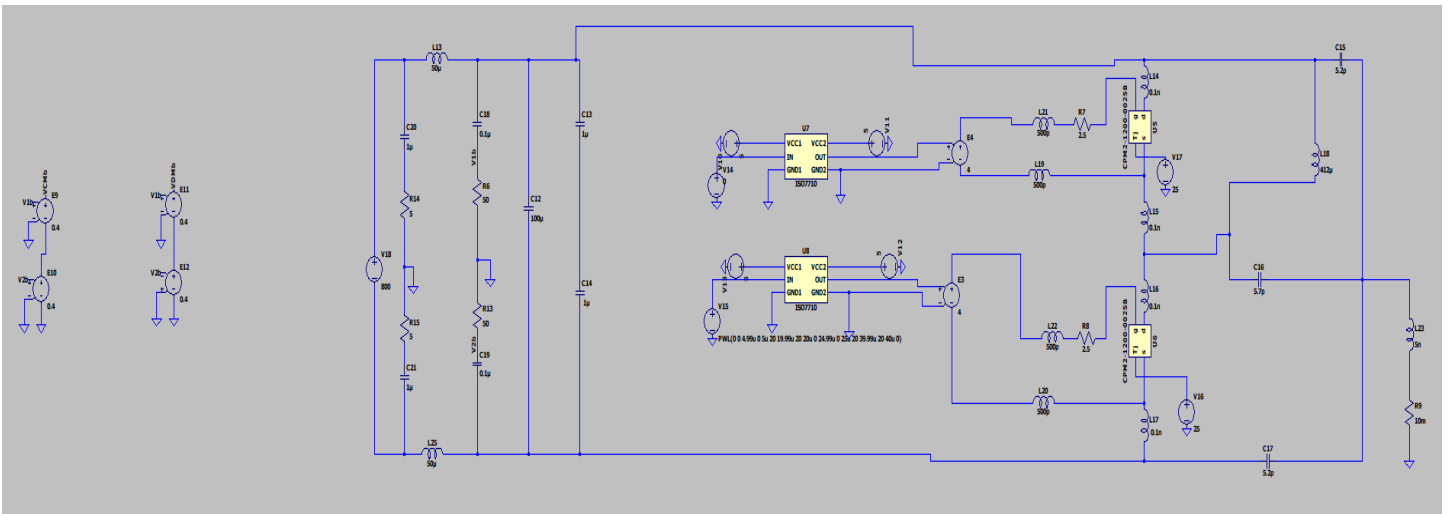


Fig2: Without Interstitial Layer

LTSPICE Simulation results

I_gnd noise without interstitial layer

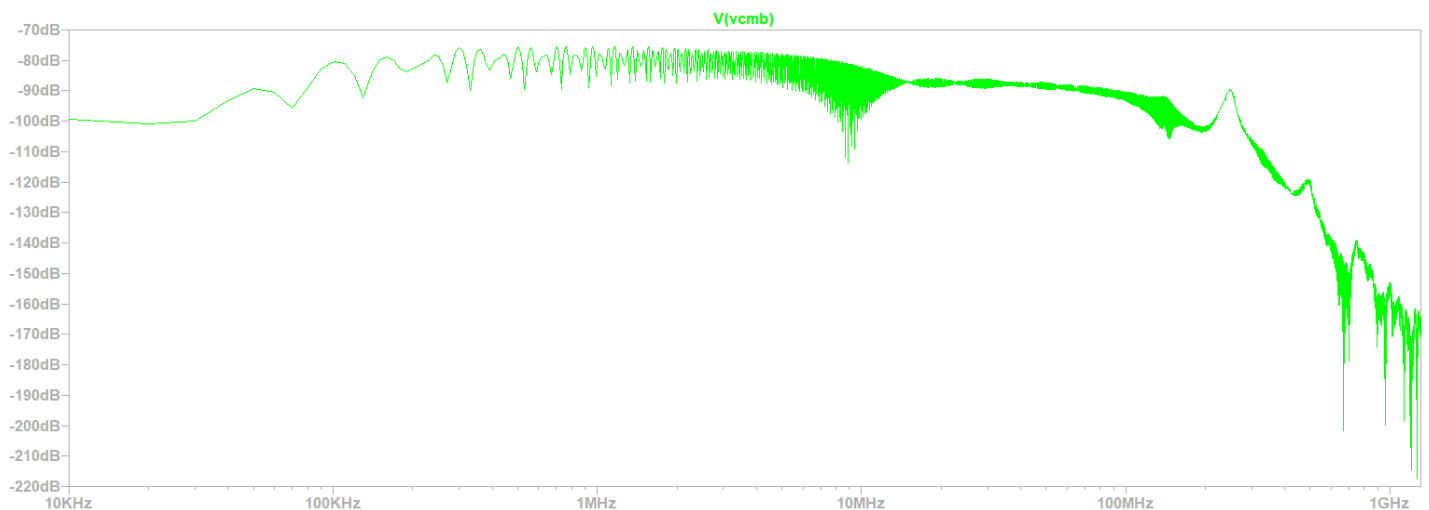


Fig3: Case A, I_gnd noise

Vds

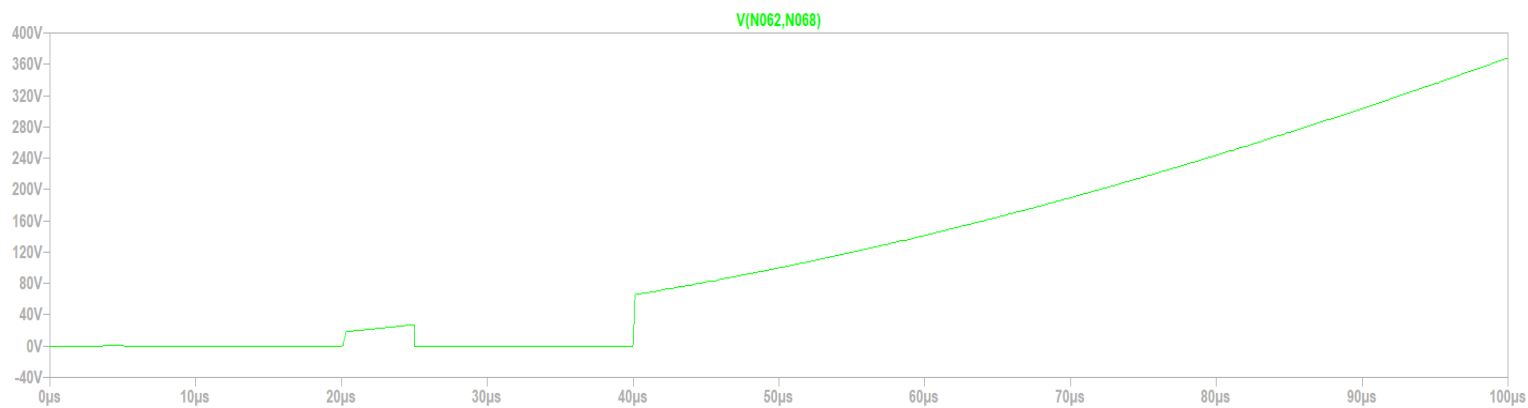


Fig4: Case A, Vds

Vgs

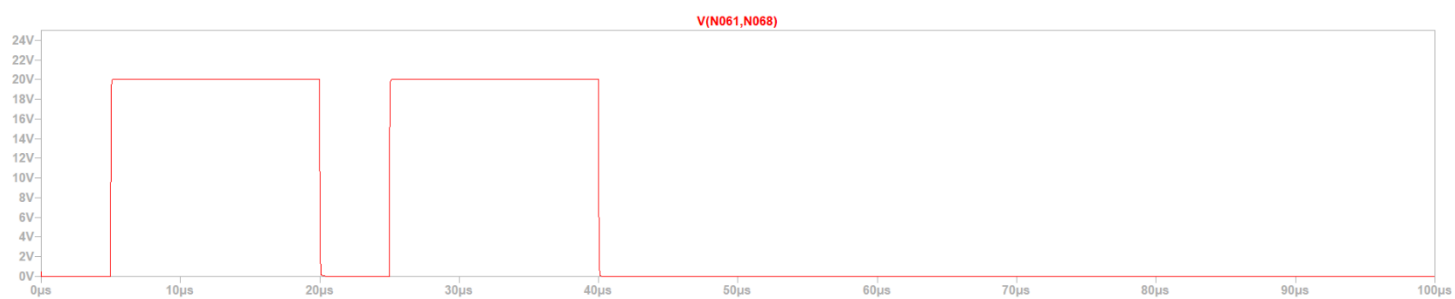


Fig5: Case A, Vgs double pulse

Ids

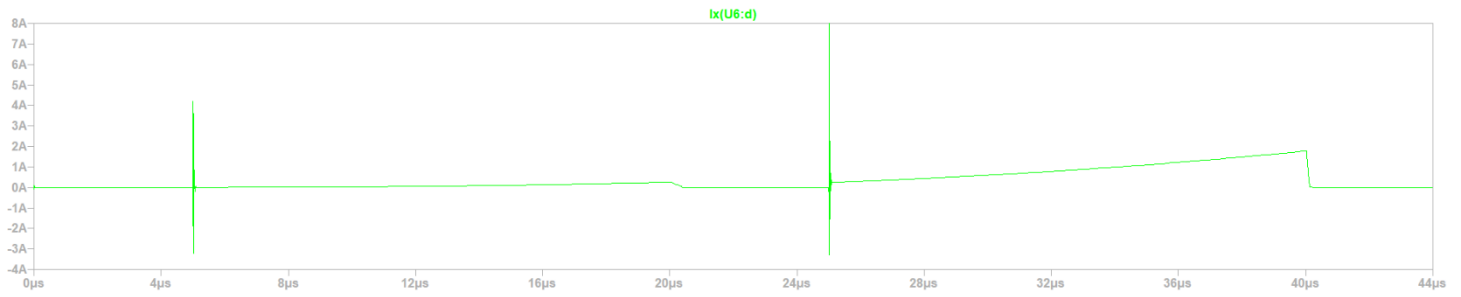


Fig6: Case A, I_{ds}

PROBLEM 1B

Case B) With interstitial layer

Gate pulse: PWL (0 0 4.99u 0 5u 20 19.99u 20 20u 0 24.99u 0 25u 20 39.99u 20 40u 0)

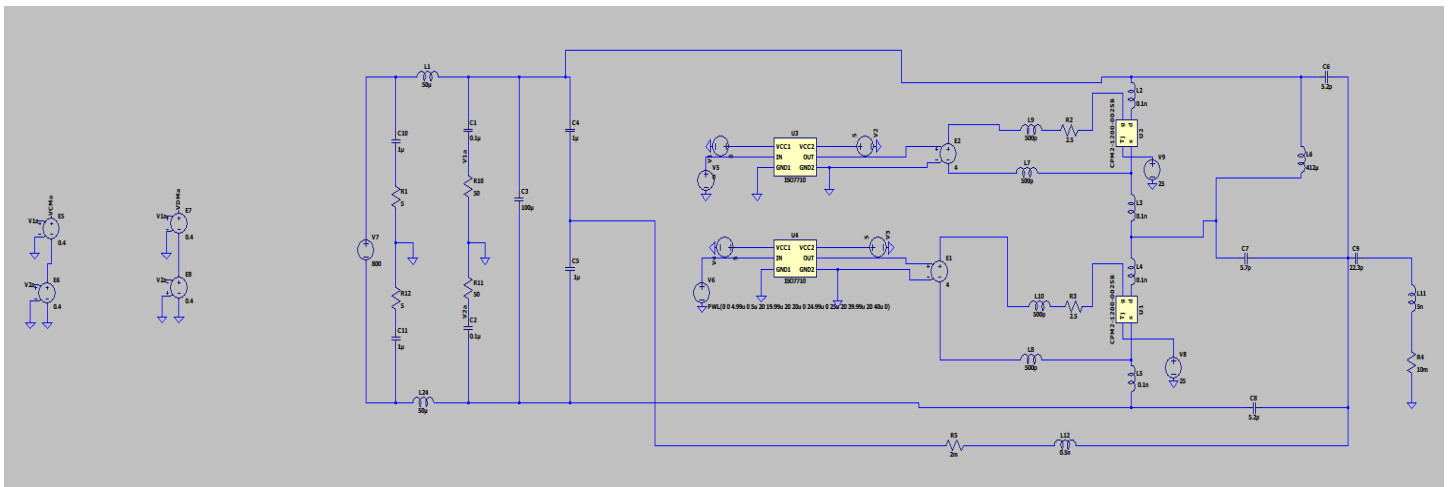


Fig7: With interstitial layer

Interstitial layer branch: $R5+L12$, associated capacitor: $C9$

LTSPICE Simulation results

I_{gnd} noise with interstitial layer

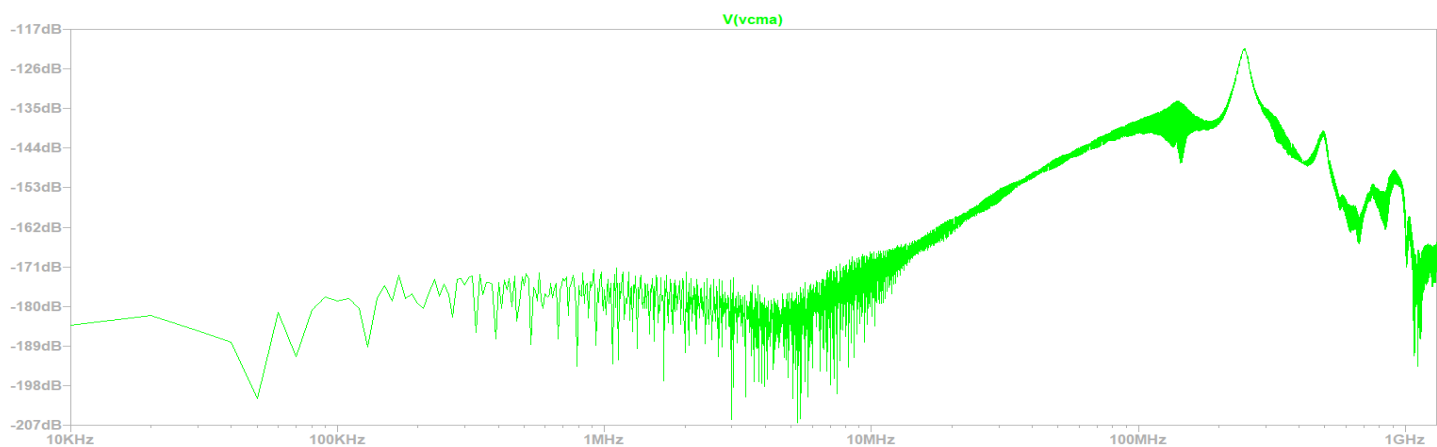


Fig8: Case B, I_gnd noise

Vds

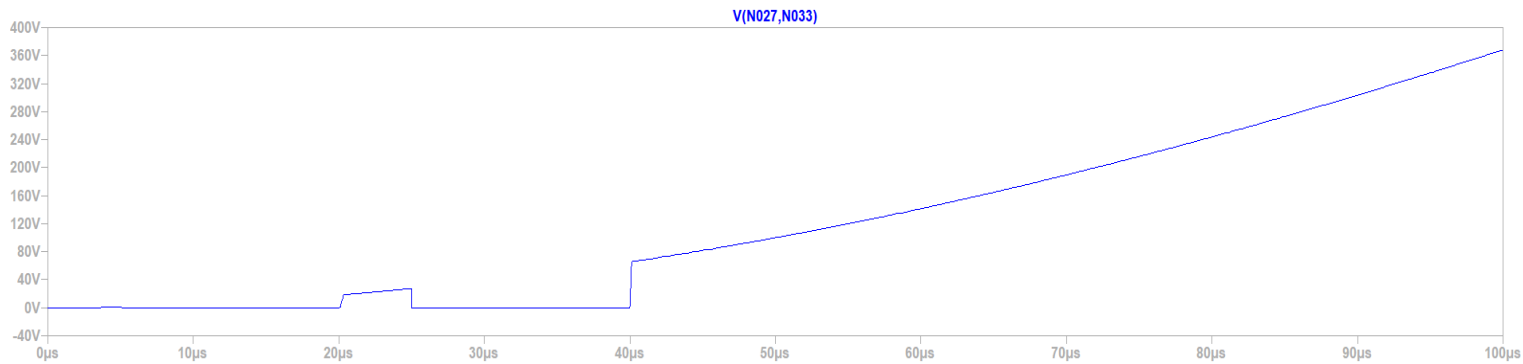


Fig9: Case B, Vds

Vgs

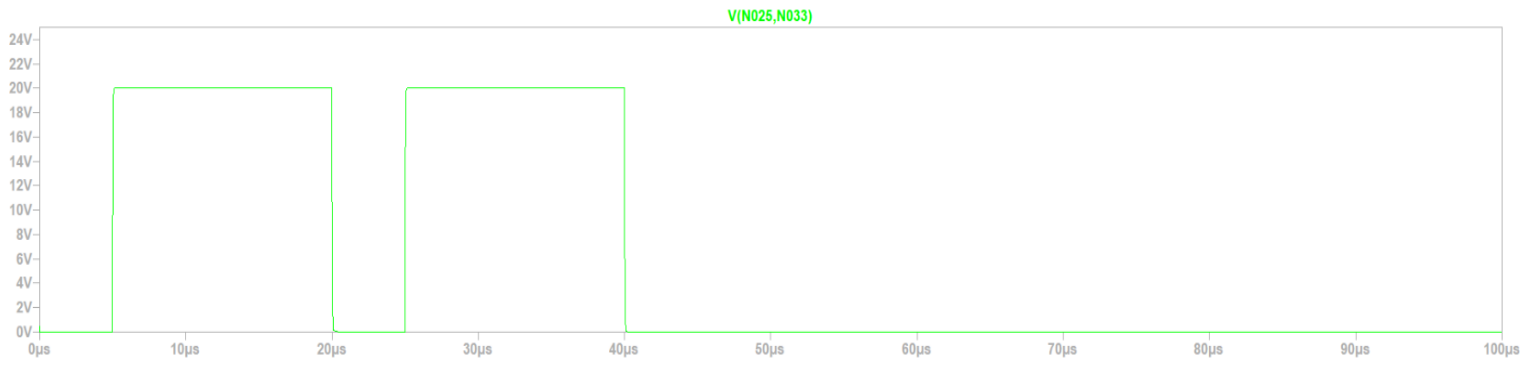


Fig10: Case B, Vgs double pulse

Ids

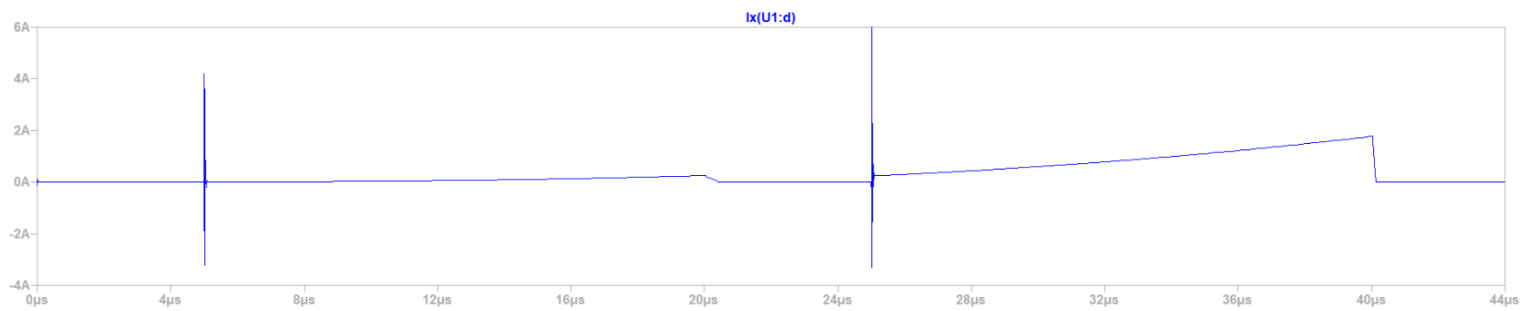


Fig11: Case B, Ids

Comparison of CaseA and CaseB I_{gnd} noise

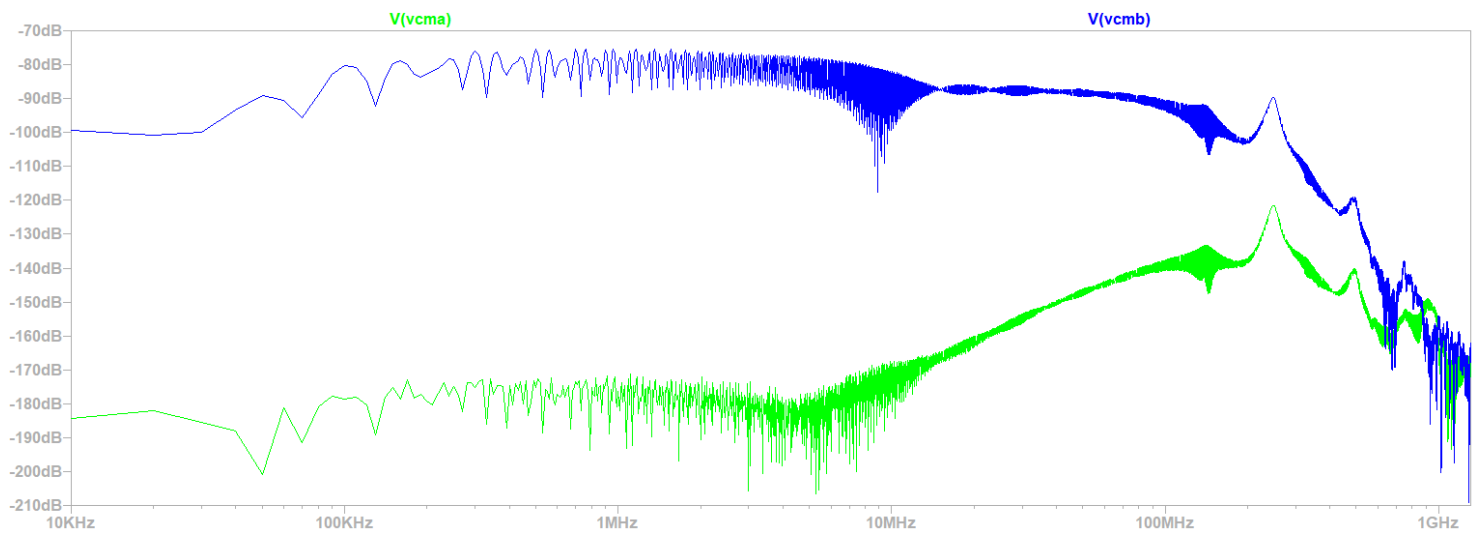


Fig12: Comparison of CaseA (blue) and CaseB (green) I_{gnd} noise

Fig12 Observation

It is clearly visible that the Case A I_{gnd} noise is higher in decibels than in Case B I_{gnd} noise for almost all the frequency till before 1GHz which directly implies that I_{gnd} in Case B is lower than in Case A. **Hence Interstitial layer benefits in reducing the offending current.**

Waveform of V_{ds} , V_{gs} and I_d in both Case A and Case B are almost same.

Problem 2: Layout of a $\frac{1}{2}$ Bridge IPM Including a Shield Layer

Introduction:

In the preliminary stage of understanding this problem, we had to understand and create a map for navigating the layout. Some of the elements used previously in the midterm were utilized to make components such as the terminals for external connections, wire bond interconnects. A new approach was implemented to integrate the gate driver and isolator into the design. These methodologies are explained extensively in the sections below.

Calculating Layer Thicknesses:

Each layer has a unique characteristic that needs to be carefully considered while dimensioning it. In the given problem, thickness for the layers 3, 4, 6 and 7 needed to be appropriately

calculated.

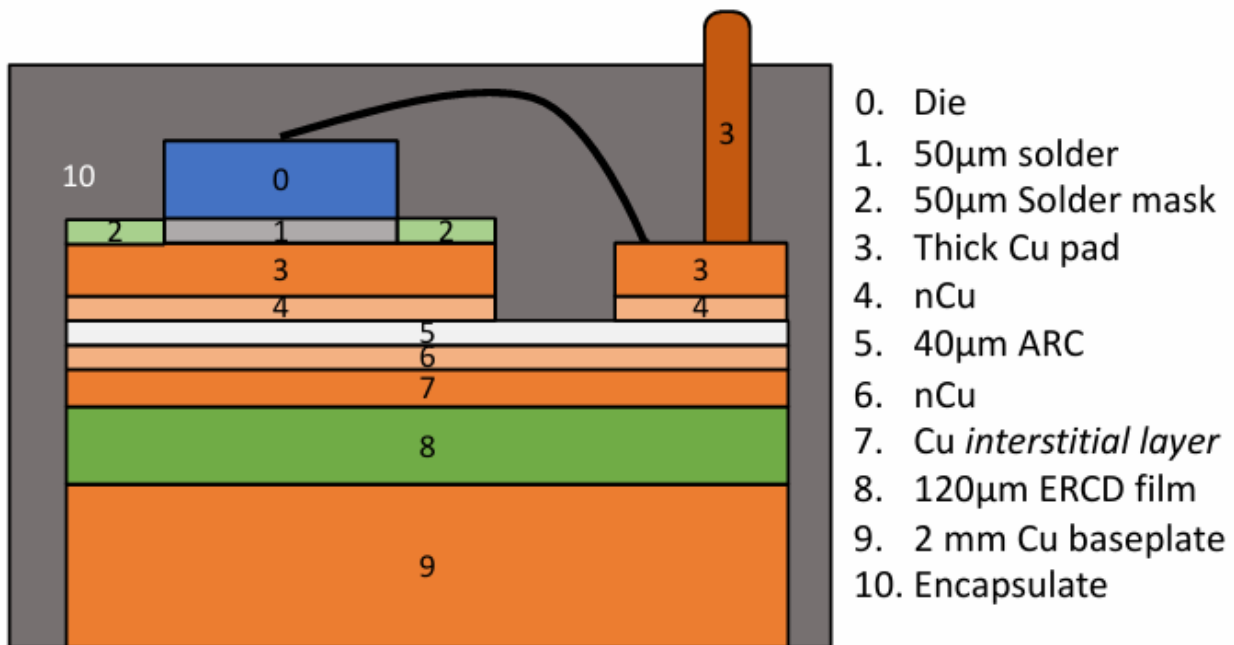


Fig13: Layer Stackup

To do this, firstly, an initial layout of all the layers was prototyped to get an idea of the X and Y dimensions, which were going to be fixed. Our first limiting variable was derived from our results in part 1 of the midterm project. In that report, we showed that the upper limit of a 10 µm nCu layer of 25-30% porosity on ribbon ceramic (ARC) was approximately 250 microns. This result was designed for an overload situation where maximum stresses would be experienced by the ARC and nCu contact without failing. This 250-micron copper thickness corresponds to a copper weight of 7oz./ft for layer 3. We utilized the IPC-2221 standard [1] for inner PCB layers to calculate the required width of the DBC pads to accommodate a 50A current flow with a 20°C temp rise limit. This width came out to 17mm. The below image shows the dimensions and spacing of the DBC layout for the power portion of the module.

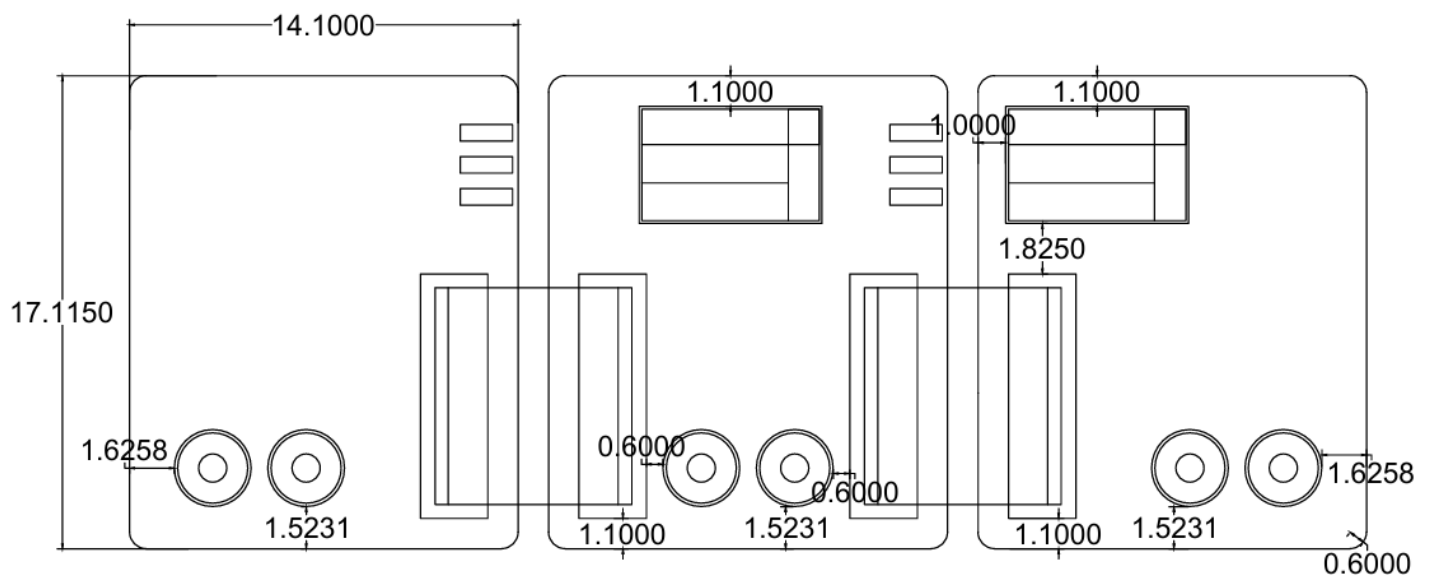
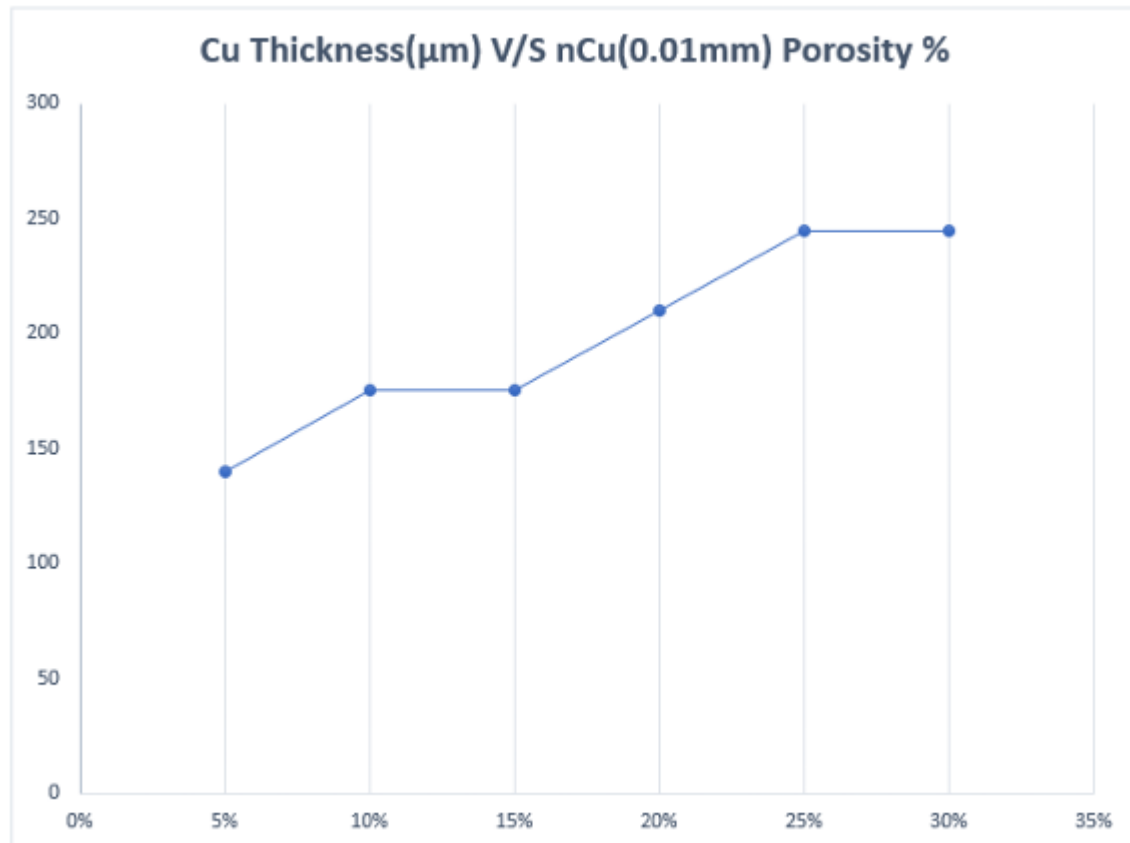


Fig14: Power Stage Dimensions

The same design rules from part 2b of the midterm project were followed here. There are no DBC pads smaller than the 40 mil limit. At least 40 mils of spacing is maintained from all edges of DBC and between all components. The corners are rounded with a radius >20 mils.

Source Wirebond size: Three wires in parallel. Nominal Current = 50A. Wire size shall be designed for higher than the current rating to handle any overload conditions, so each wire should be rated for at least 30A. From the table below: 300 μ m diameter, no more than 7.5mm length

Gate Wirebond Size: Gate wires are sized for 2.4A, so 50 μ m diameter, 5mm max length.

The layouts are designed to maintain the wirebond length limits. In addition, the pad Widths of WolfSpeed MOSFET are large enough for the 12mil wire. The gate pad width is similarly large enough for 2 mil wire (not shown in chart). Appropriately sized bond pads are indicated in Fig 14.

Wire Dia μ m (mils)	A1 (99.99%)						Al-1% Si					
	Wire length (mm)						Wire length (mm)					
	1.3	2.5	5	7.5	10	12.5	1.3	2.5	5	7.5	10	12.5
25 (1)	1.2	0.6	0.3	0.2	0.2	0.1	0.8	0.4	0.2	0.1	0.1	0.1
50 (2)	4.8	2.4	1.2	0.8	0.6	0.5	3.2	1.6	0.8	0.5	0.4	0.3
75 (3)	11	5.4	2.7	1.8	1.4	1.1	7.2	3.6	1.8	1.2	0.9	0.7
100 (4)	19	9.6	4.8	3.2	2.4	1.9	13	6.4	3.2	2.1	1.6	1.3
125 (5)	30	15	7.5	5.0	3.8	3.0	20	10	5.0	3.3	2.5	2.0
200 (8)	38	19	13	9.6	7.7		26	13	8.5	6.4	5.1	
250 (10)	60	30	20	15	12		40	20	13	10	8.0	
300 (12)	86	43	29	22	17		58	29	19	14	12	
375 (15)	135	68	45	34	27			45	30	23	18	
450 (18)	194	97	65	49	39			65	43	32	26	
500 (20)	240	120	80	60	48			80	53	40	32	
625 (25)	375	188	125	94	75			125	83	63	50	

Maximum-Useful Bond Pad dimensions					
Wire Size		Pad Width (W)		Pad Length (L+T)	
mil	μ m	mil	μ m	mil	μ m
4	100	7.7	200	24	610
5	125	9.3	240	29	750
6	150	11	280	33	840
7	175	13	320	40	1010
8	200	14	370	44	1130
10	250	19	490	56	1410
12	300	23	580	67	1710
14	350	27	680	77	1960
15	375	28	720	83	2110
16	400	30	770	88	2240
18	450	34	860	99	2520
20	500	38	960	110	2800

Fig15a: Bond Wire Fusing Currents Fig15b: Bond Wire Bad Dimensions

The last portion of layer 3 are the terminals, which should also be rated for 50A. In this design, we have specified Terminal: 1440-1-00-15-00-00-03-0. In keeping with the spirit of a productized HBPM, we used the WolfSpeed CAB004M12GM4T half bridge power module [2] as a reference. This product features terminals of similar cross-sectional area as our chosen terminals, and WolfSpeed rates them for 25A each. We have included two terminals for each connection point to satisfy the 50A requirements.

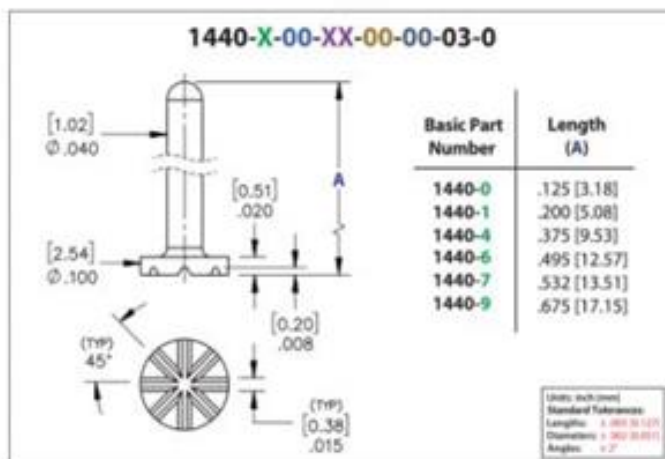


Fig16: Terminal Specifications

Other properties, such as equivalent resistances and parasitics, were used to determine the thickness of the remaining layers. Some of the results were derived from previously developed solutions, for e.g.: the currents flowing through the interstitial layers 6 and 7 and those flowing via layers 3 and 4. Layers 6 and 7 collectively are shielding layers. The R-L branch of the equivalent circuit model that creates feedback to the circuit is representative of these layers. A small current in the order of mA flows back to the circuit. This current was considered to calculate the thicknesses required. The layers from 5 onwards formed a base of a constant area, upon which the rest of the copper layers, dyes and interconnects were then appropriately placed. Given below is a table of the widths of the layers required.

Layer	Thickness (μm)
Thick Cu Pad	246.3
NCu - 1	10
NCu - 2	10
Interstitial Cu	59.6

The isolation and gate drive circuitry was implemented in Altium and 3D files were exported to be combined with the HBPM power stage in SolidWorks. Hence, PCB design rules were followed instead of DBC rules. However, the gate driver copper has identical thicknesses to layers 3 and 4 of the DBM power stage to maintain structural compatibility with the ARC dielectric. One important note for the isolation layout is that there is no copper directly underneath the isolation IC. Some leeway was taken in assuming that the gate drive circuit is manufacturable as depicted. In reality, it would likely be a separate PCB that is integrated into the same power module and encapsulated. Figure 17 shows the schematic for one of the two isolated gate drive circuits. The inputs required are an isolated DC power supply (like from a battery) across pins Vbatt and GND of the connector. Also needed is the gate drive PWM signal. To maintain consistency with the producibility of this module, an off-the-shelf connector was selected for this signal and low voltage pins.

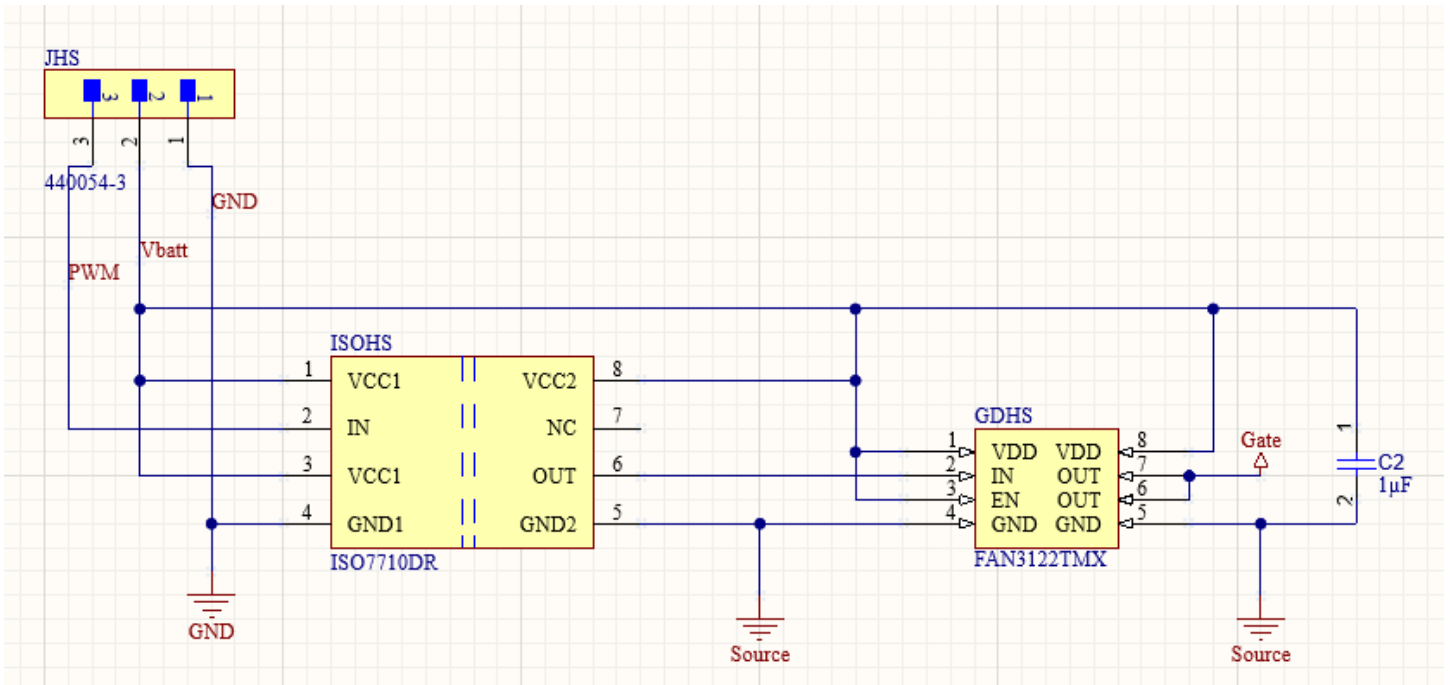
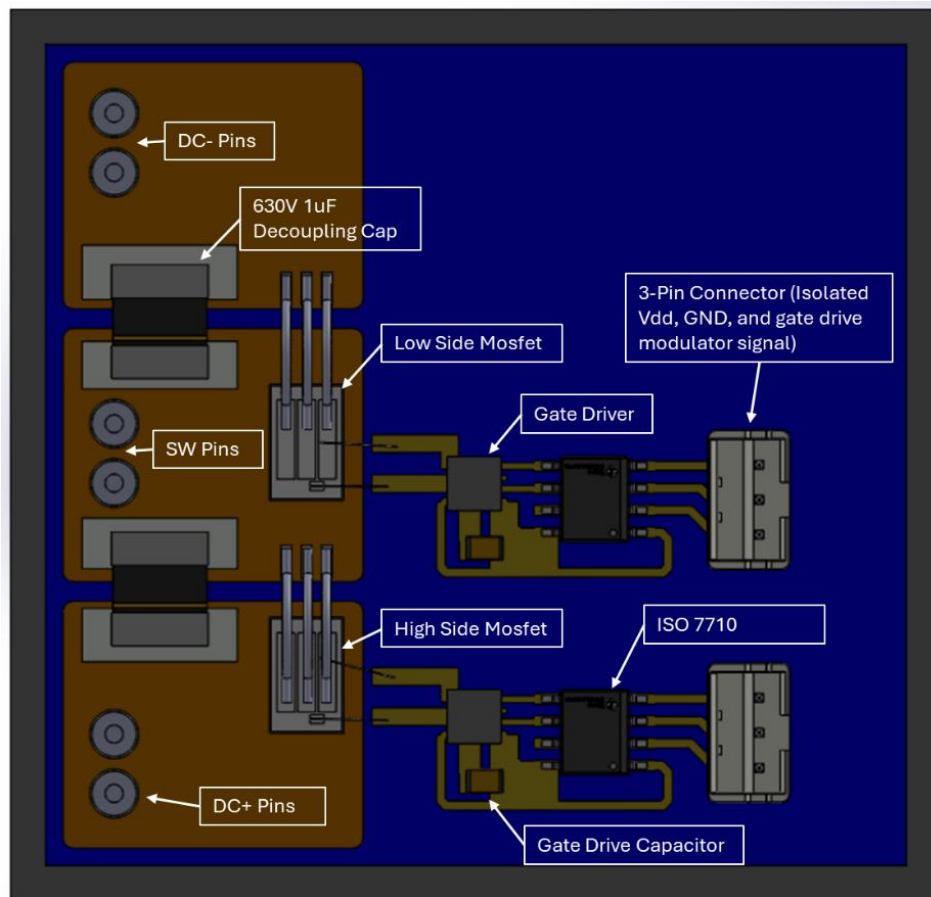


Fig17: Isolated Gate Drive Circuit

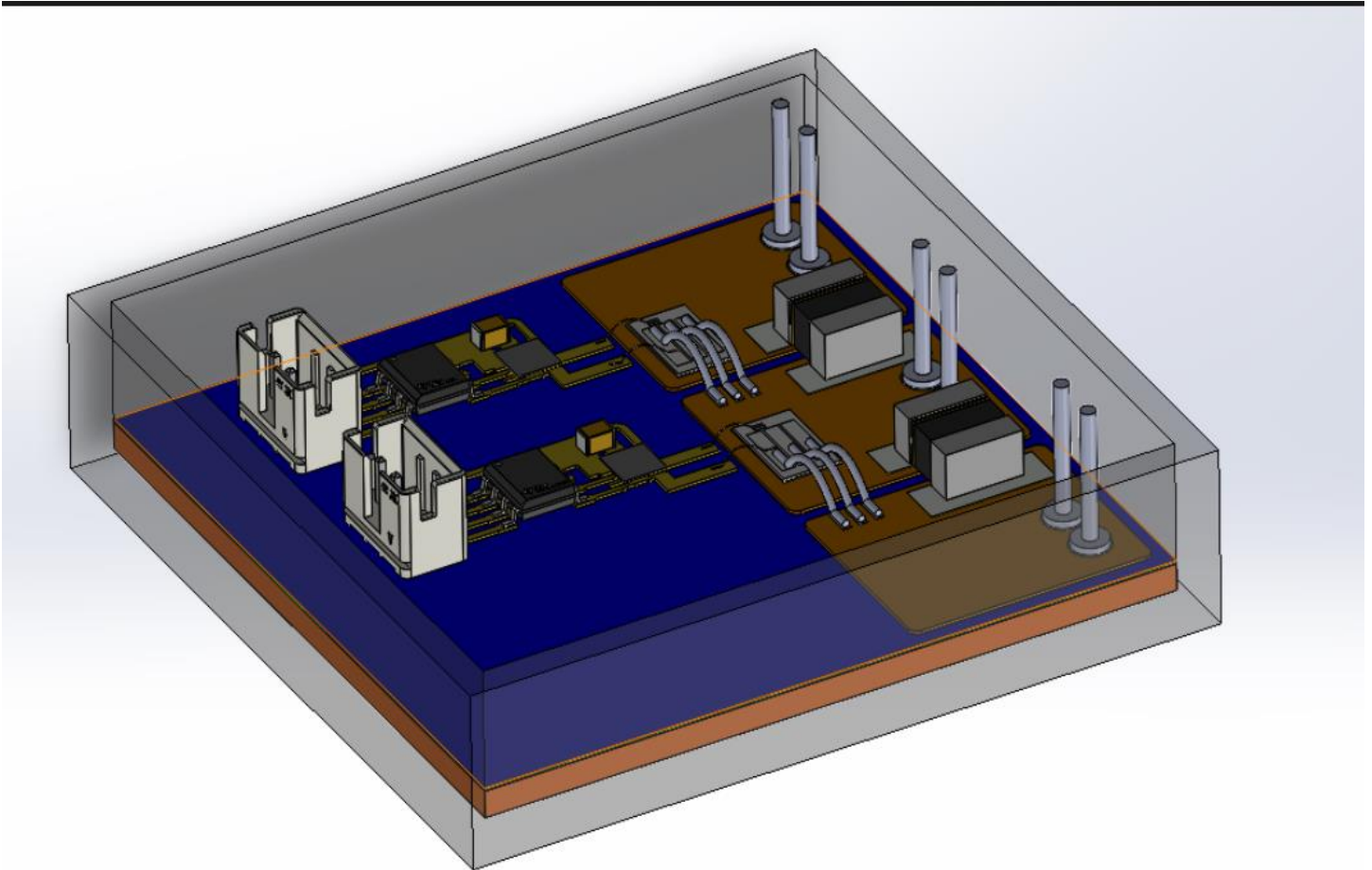
In the image below, to the left is the HB module with the gate driver and isolator to the right. A 630V, 1μF decoupling capacitor was selected for the MOSFETs. A CAD model of the FAN3122 gate driver was not available, hence, the block shown is representative. The two pins in the middle are the HB outputs, made for handling higher ampacity. The lower and upper parts are the high side and low side parts of the bridge respectively.



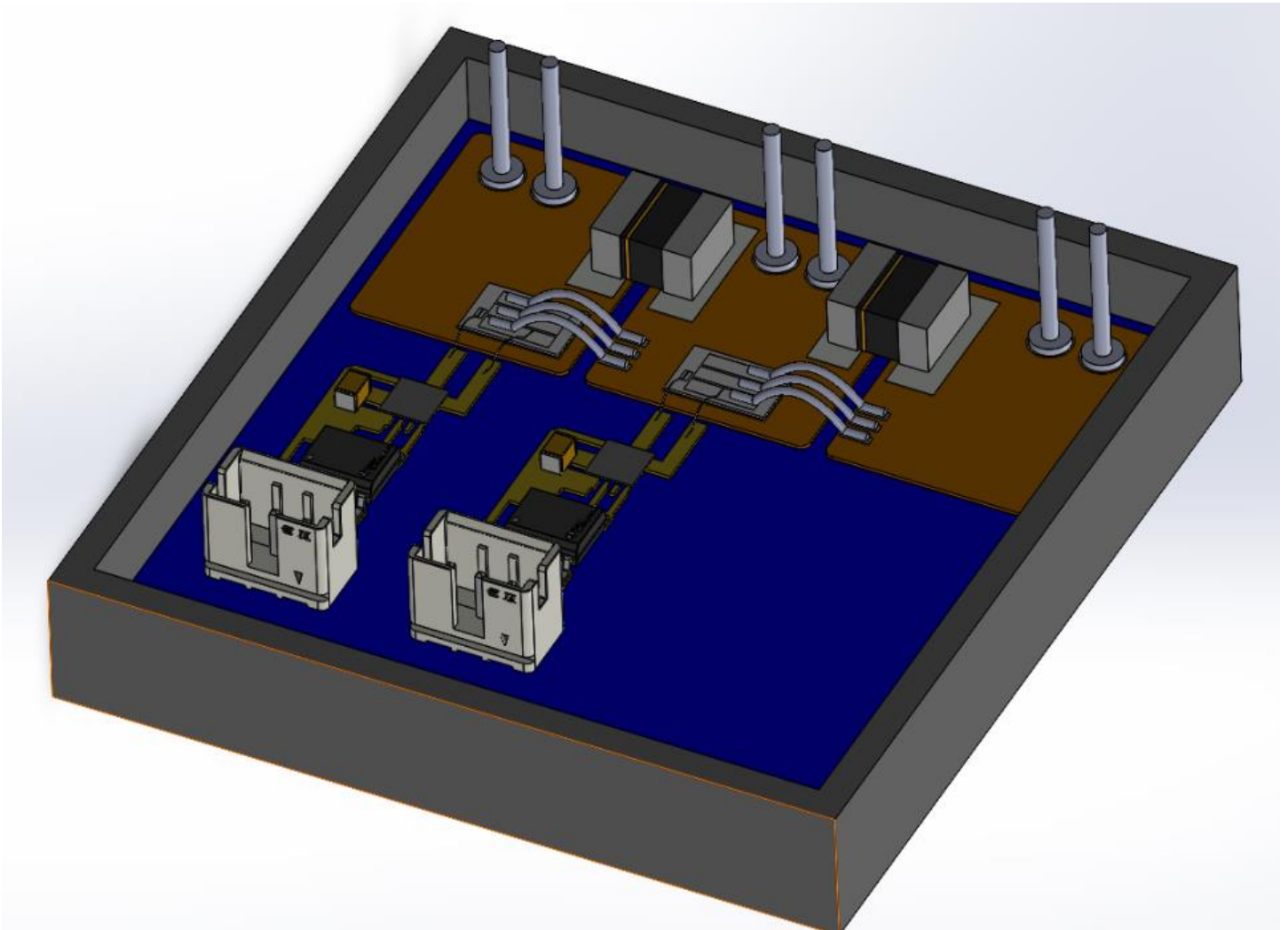
Top View without Encapsulation

Design Methodology and Assumptions:

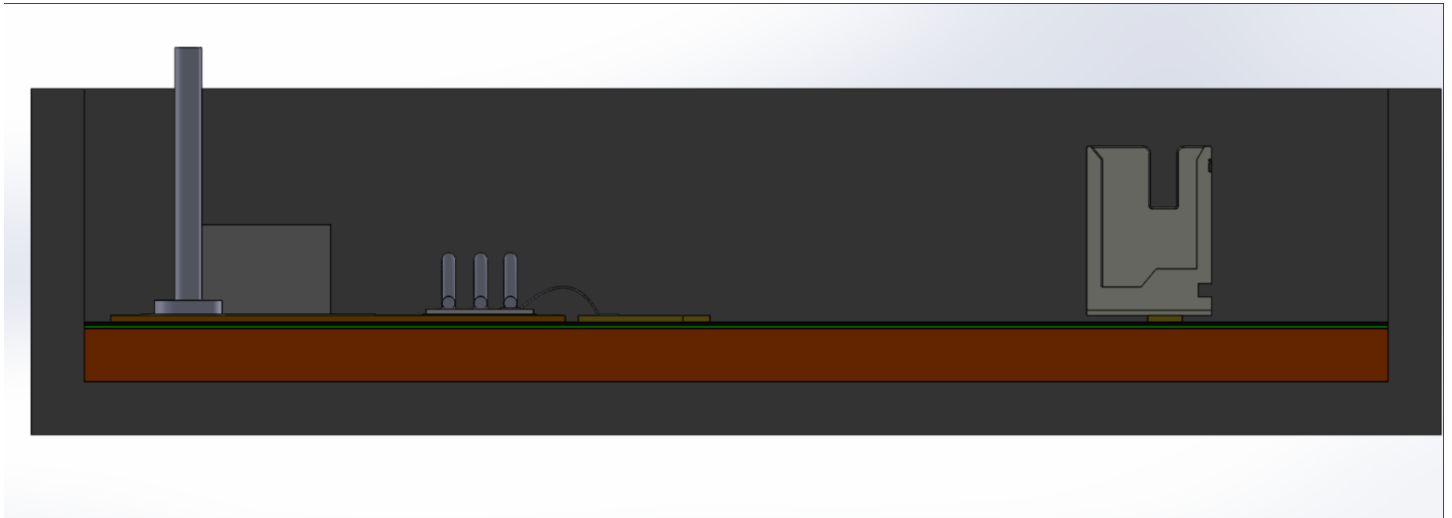
Final Design:



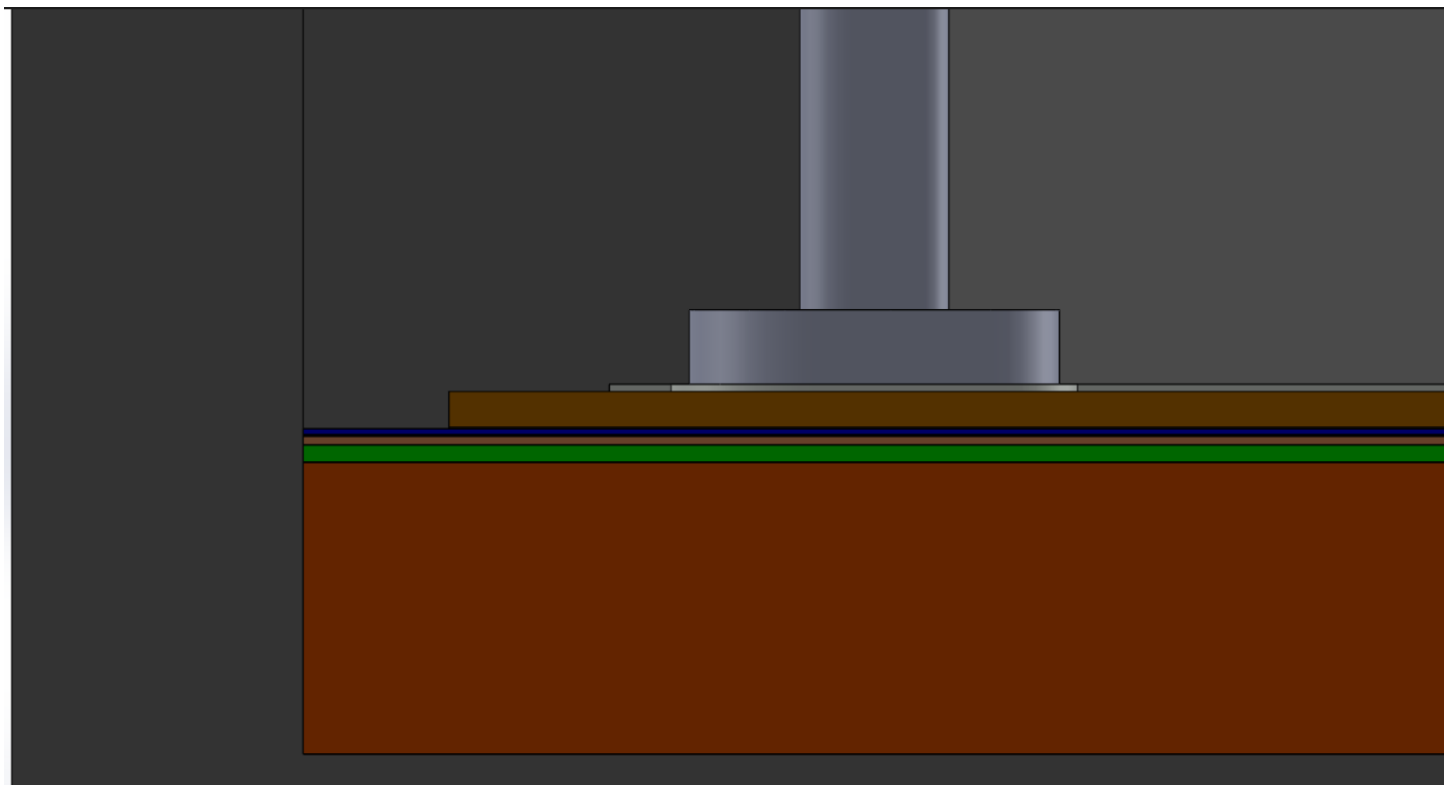
Transparent Encapsulation View of the Complete Unit



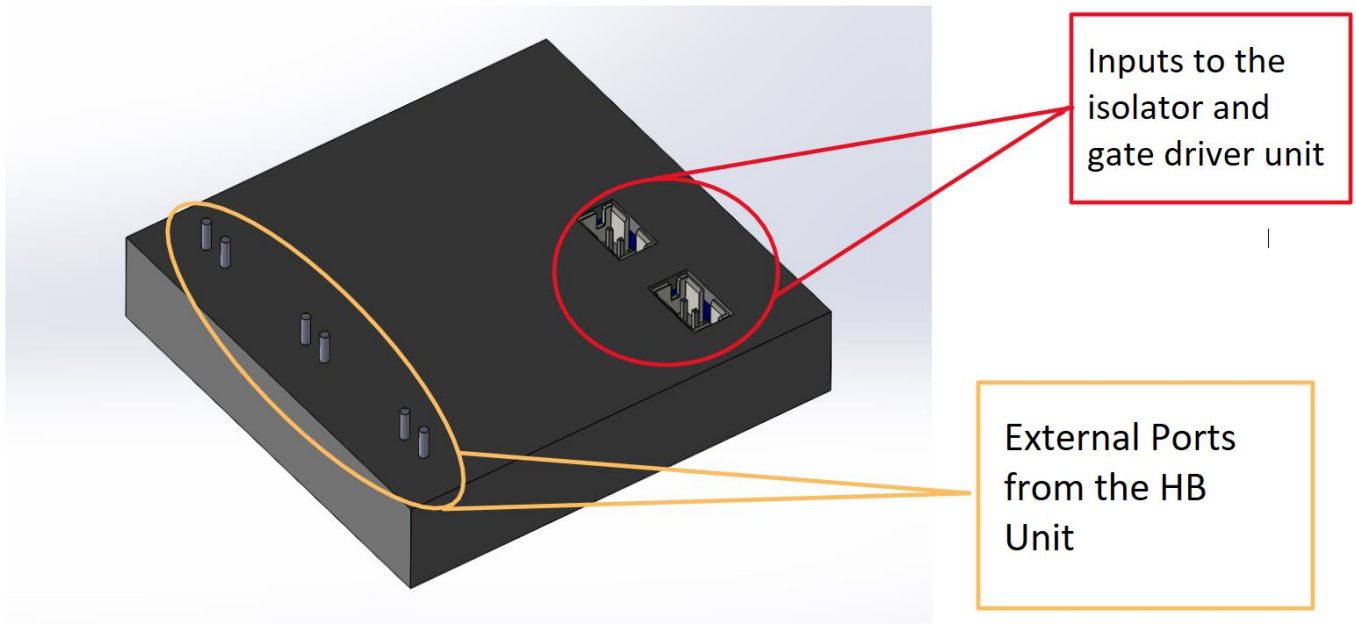
Complete Model View without Encapsulation Top



Front View with Transparent Encapsulation



Zoomed In Front View



Complete IPM Model with Encapsulation

References:

- (1) <https://www.advancedpcb.com/en-us/tools/trace-width-calculator/>
- (2) [CAB004M12GM4T, 1200V Half-Bridge Power Module | Wolfspeed](#)